

Author: Pankaj Yadav

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Description:

This assignment is to learn connecting GenAI API and using Retrieval Augmented Generation (RAG).

Instructions

Create a chat like the one in this week's lecture video using any Wikipedia page except the one in the video. Be sure to ask your bot at least two separate questions.

Build the solution using a Jupyter Notebook (no other Python is acceptable) and include Markdown that thoroughly explains your thought process and your commentary on the results you achieved. Obvious comments that explicitly only explain the code will be ignored since I can read the code. I'm looking for your commentary and evaluation in your own words.

I setup os, chromadb, pprint, operator, and OpenAI-related libraries for this notebook workflow. os.getenv helps me fetch the API key from environment variables instead of hardcoding it, pprint helps display outputs in a clean readable way, and index is used while creating unique IDs for stored document chunks. WikipediaLoader is used to pull source content, RecursiveCharacterTextSplitter breaks it into smaller chunks, Chroma and chromadb store and retrieve embeddings, OpenAIEmbeddings converts text into vectors, PromptTemplate formats the question context, ChatOpenAI connects to the model for responses, and RetrievalQA combines retrieval plus LLM answering in one chain.

```
In [113... import os
import chromadb
import pprint
from operator import index
from openai import OpenAI
from langchain_community.document_loaders import WikipediaLoader
from langchain_text_splitters import RecursiveCharacterTextSplitter
from langchain_community.vectorstores import Chroma
from langchain_openai import OpenAIEmbeddings
from langchain_classic.chains import RetrievalQA
from langchain_core.prompts import PromptTemplate
from langchain_openai.chat_models import ChatOpenAI
```

I set the search topic to IPL 2026 which is the cricket tournament and used WikipediaLoader to pull content directly from Wikipedia so the chatbot can answer questions from a real, up-to-date source. The query uses load_max_docs=1 to focus on one main article and avoid mixing unrelated pages, while doc_content_chars_max=300000 allows a much larger portion of the page to be captured. This larger limit is important because match-by-match details are usually listed much lower in the article, and a small character cap can miss those sections.

```
In [114... search_term = "IPL 2026"
docs = WikipediaLoader(query=search_term, load_max_docs=1, doc_content_chars_max=300000).load()
```

I use RecursiveCharacterTextSplitter to break the Wikipedia article into smaller, meaningful chunks before creating embeddings. I set chunk_size to 1000 so each piece is small enough for accurate retrieval, and chunk_overlap to 20 so important details that fall near boundaries are not lost between chunks. length_function=len measures chunk size by characters, and is_separator_regex=False keeps splitting behavior simple and stable using normal separators. This setup improves retrieval quality for specific match-level questions in the RAG pipeline.

```
In [115... text_splitter = RecursiveCharacterTextSplitter(chunk_size=1000
                                                , chunk_overlap = 20
                                                , length_function = len
                                                , is_separator_regex=False, )
```

In this step, I split the loaded Wikipedia content into smaller document chunks using the text splitter configuration defined earlier, and store those chunks in data. Running data right after that helps me quickly inspect the output structure, confirm chunking worked correctly, and verify that important sections from the source page are present before moving to embeddings and vector storage. This checkpoint is useful for validating retrieval quality early in the RAG pipeline.

```
In [116... data = text_splitter.split_documents(docs)
data
```

Out[116... [Document(metadata={'title': '2026 Indian Premier League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.', 'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League'}, page_content='The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.\n\n\n== Background ==\n\nThe Indian Premier League (IPL) is a professional Twenty20 (T20) cricket league, organized by the Board of Control for Cricket in India (BCCI). It has held annually since the first edition in 2008. Royal Challengers Bengaluru are the defending champions, having won their maiden title in the previous season after defeating Punjab Kings in the final.'),

Document(metadata={'title': '2026 Indian Premier League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.', 'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League'}, page_content='=== Format ===\n\nThe group format has been reversed to be similar to that of 2023 IPL. Each team will play once against the teams in their group and twice against the teams in the other group. After the group stage, the top four teams, based on aggregate points, would advance to the playoffs. In this stage, the top two teams compete with each other (in a match titled "Qualifier 1"), as do the remaining two teams (in a match titled "Eliminator"). While the winner of Qualifier 1 would directly qualify for the final match, the losing team had another chance to qualify for the final match by competing against the winning team of the Eliminator (in a match titled "Qualifier 2"). The winner of this subsequent Qualifier 2 match advanced to the final match. Teams were seeded into groups based on the number of titles they have won.'),

Document(metadata={'title': '2026 Indian Premier League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.', 'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League'}, page_content='The IPL Governing Council had initially announced that the IPL would expand to 84 matches for 2026 and 2027, with it expected to expand to 94 matches from 2028 onwards with the return of the complete double round-robin format that was used until 2021. However, the number of matches was kept as 74 for one more edition as it was in the previous four seasons.'),

Document(metadata={'title': '2026 Indian Premier League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.', 'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League'}, page_content='=== Schedule ===\n\nAccording to ESPNcricinfo, the dates of the tournament window for the three seasons between 2025 and 2027 were sent to franchises ahead of the 2025 auction. In November 2024, the tentative dates for the three seasons were announced, with the 2026 edition set to take place from 15 March to 31 May. In December 2025, it was announced that the 2025 season would commence on 26 March to allow for a three-week gap after the 2026 Men's T20 World Cup final scheduled to take place on 8 March. In February 2026, it was announced that the 2026 season would commence 2 days later on 28 March. This led to a clash with the 2026 Pakistan Super League for the second consecutive year taking place from 26 March to 3 May 2026, which was again moved from its typical February–March window for the same reason.'),

Document(metadata={'title': '2026 Indian Premier League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.', 'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League'}, page_content='In March 2026, it was announced that the fixtures for that season would be announced in two parts due to the State Assembly Elections in Assam, Tamil Nadu and West Bengal; for which the dates were not announced till 15 March. The first part of the schedule comprising the first twenty matches was announced on 11 March, with the inaugural match being hosted at M. Chinnaswamy Stadium between the defending champions Royal Challengers Bengaluru and Sunrisers Hyderabad on 28 March. The complete fixtures for the league stage were announced on 26 March. The playoffs schedule is yet to be announced.'),

Document(metadata={'title': '2026 Indian Premier League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.', 'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League'}, page_content='=== Exclusion of Mustafizur Rahman ==='),

Document(metadata={'title': '2026 Indian Premier League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.', 'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League'}, page_content="Bangladeshi fast bowler Mustafizur Rahman was bought by Kolkata Knight Riders for ₹9.2 crore (US\$1.1 million) in the auction. Following anti-Hindu violence in Bangladesh during 2025 and the lynching of Dipu Chandra Das in December 2025, there were calls for the exclusion of Bangladeshi players from the IPL. In January 2026, the BCCI requested that the franchise release Rahman from the squad and permitted them to select a replacement player. This move was criticized by former Indian cricketer Madan Lal, Congress politician Shashi Tharoor, and former Bangladeshi captains Khaled Mahmud and Mohammad Ashraful while former Indian cricketer Aakash Chopra and BJP politician Sangeet Singh Som defended the BCCI's move."),

Document(metadata={'title': '2026 Indian Premier League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.', 'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League'}, page_content="This led the BCCI to put the Indian tour of Bangladesh on hold due to tensions. Furthermore, the Bangladesh Cricket Board requested the International Cricket Council (ICC) to shift their matches at the 2026 Men's T20 World Cup which was to be co-hosted by India and Sri Lanka, out of India, similar to that of Pakistan's. But the request was rejected by the ICC, and Bangladesh ultimately withdrew from the T20 World Cup. The Bangladesh government also banned the broadcast of IPL in their country. Following the 2026 Bangladeshi general election, the newly elected government clarified that the broadcast of the IPL is not banned anymore in their country."),

Document(metadata={'title': '2026 Indian Premier League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.', 'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League'}, page_content="=== Marketing ===\n\nThe Tata Group renewed their contract as the title sponsors of the IPL for a tenure of 5 years (2024–2028) for ₹2,500 crore (US\$300 million).\n\n\n=== Broadcasting ===\n\nJioStar's Star Sports and JioHotstar, currently holds the satellite and digital broadcasting rights for the remainder of the 2023–2027 cycle respectively.\n\n\n\n=== Teams ===\n\nThe same 10 teams from the previous season returned.\n\n\n\n=== Personnel changes ==="),

anges ==="),

Document(metadata={'title': '2026 Indian Premier League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.', 'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League'}, page_content='The franchises were required to submit their retention lists before 15 November 2025, and a total of 173 players were retained ahead of the auction. The auction was held on 16 December 2025 in Abu Dhabi, United Arab Emirates. with the auction purse for each franchise set at ₹125 crore (US\$15 million). A total of 1,355 players registered for the auction, of which 369 players were shortlisted to feature in the auction and 77 were sold in the auction. Cameron Green became the most expensive overseas player and the third most expensive player in the history of IPL when he was bought by Kolkata Knight Riders for ₹25.20 crore (US\$3.0 million). Prashant Veer and Kartik Sharma became the joint-most expensive uncapped players in the history of IPL when they were both bought by Chennai Super Kings for ₹14.20 crore (US\$1.7 million) each. Meanwhile the 79 unsold players notably included Jake Fraser-McGurk, Jonny Bairstow, Daryl Mitchell, Devon Conway, Alzarri Joseph, Gus Atkinson, Deepak Hooda'),

Document(metadata={'title': '2026 Indian Premier League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.', 'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League'}, page_content='Deepak Hooda and Karn Sharma.'),

Document(metadata={'title': '2026 Indian Premier League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.', 'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League'}, page_content='== Venues ==\n\nThe league stage is being played at 13 stadiums across India. The opening match was played at the M. Chinnaswamy Stadium. The HPCA Cricket Stadium will host three Punjab Kings home matches, the ACA Cricket Stadium will host three Rajasthan Royals matches, and the Nava Raipur Cricket Stadium will host two Royal Challengers Bengaluru matches. The playoffs\' venues are yet to be confirmed.\n\n\n\n== League stage ==\n\n\n\n=== Points table ===\n\n\n\n=== Match summary ===\n\n\n\n=== Fixtures ===\n\n\n\n== Playoffs ==\n\n\n\n=== Bracket ===\n\n\n\nSource: ESPNcricinfo\n\n\n\n=== Qualifier 1 ===\n\n\n\n=== Eliminator ===\n\n\n\n=== Qualifier 2 ===\n\n\n\n=== Final ===\n\n\n\n== Statistics ==\n\n\n\n== References ==\n\n\n\n=== General references ===\n\n\n\nIPL Season Schedule 2026 Part 1" (PDF). Indian Premier League Schedule. 19 (1). Board of Control for Cricket in India. 11 March 2026.\n\n\nIPL Season Schedule 2026 Complete" (PDF). Indian Premier League Schedule. 19 (2). Board of Control for Cricket in India. 26 March 2026.'),

Document(metadata={'title': '2026 Indian Premier League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.', 'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League'}, page_content='== External links ==\n\nOfficial website\n\n2026 Indian Premier League at ESPNcricinfo'))]

In this step, I securely load the OpenAI API key from environment variables using `os.getenv` instead of exposing it directly in the notebook. Then I initialize `OpenAIEmbeddings` with that key so each text chunk can be converted into vector embeddings for semantic search in the Chroma database. This is a core RAG step because retrieval quality depends on how well these embeddings represent the meaning of the source content.

```
In [117... api_key = os.getenv("OPENAI_API_KEY")
embeddings = OpenAIEmbeddings(api_key=api_key)
```

In this step, I create the Chroma vector store from the chunked Wikipedia data and generated embeddings so the chatbot can retrieve relevant context during question answering. I also assign custom IDs using each chunk's source metadata plus its index to keep records unique and traceable. The collection is named IPL_2026 for clear organization, and persist_directory='db' saves the vector database locally so I can reuse it later without rebuilding embeddings every time I run the notebook.

```
In [118... store = Chroma.from_documents(
    documents=data,
    embedding=embeddings,
    ids = [f"{item.metadata['source']}-{index}" for index, item in enumerate(data)],
    collection_name="IPL_2026",
    persist_directory='db',
)
```

In this step, I define the prompt template that controls how the LLM should answer questions in the RAG flow. The template tells the model to rely on retrieved context, answer the user question directly, and return "I don't know" when the information is missing, which helps reduce hallucinations. The placeholders {context} and {question} are dynamically filled at runtime with retrieved chunks and the user query, so the response stays grounded in the indexed Wikipedia content.

```
In [119... template = """You are the bot that answers questions about the Indian Premier League (IPL) 2026.
Use the following pieces of context to answer the question at the end. If you don't know
the answer, say you don't know. Always use all available data to answer the question.
{context}
Question: {question}"""
```

In this step, I convert the raw prompt text into a reusable PromptTemplate object so it can be injected cleanly into the RetrievalQA chain. The input variables context and question map to the retrieved chunks and the user query at runtime, ensuring each response is generated using both relevant evidence and the exact question asked. This makes the prompt dynamic, structured, and consistent across all chatbot queries.

```
In [120... prompt = PromptTemplate(
    template = template,
    input_variables = ["context", "question"],
)
```

In this step, I initialize the chat language model that will generate final answers from the retrieved IPL context. I set temperature=0 to keep responses more deterministic and factual, which is important for RAG tasks where consistency matters. The api_key is passed securely from environment variables so the notebook can authenticate with OpenAI without hardcoding secrets.

```
In [121... llm = ChatOpenAI(model="gpt-5.4", temperature=0, api_key=api_key)
```

In this step, I build the final RetrievalQA chain that connects retrieval and generation into one pipeline. The retriever pulls the most relevant chunks from the Chroma store, and the llm uses the custom prompt to answer the question using that context. I chose chain_type as stuff so the retrieved chunks are combined and sent together to the model, and return_source_documents=True helps me inspect which source chunks were used, making the output more transparent and easier to validate.

```
In [122... qa_with_context = RetrievalQA.from_chain_type(
    llm=llm,
    retriever=store.as_retriever(),
    chain_type="stuff",
    chain_type_kwargs={"prompt": prompt,},
    return_source_documents=True,
)
```

In this step, I run the first user query through the complete RAG pipeline to test whether the chatbot can retrieve relevant IPL 2026 context and produce a grounded answer. The question checks a basic factual detail ("When did IPL 2026 start?"), which is a good sanity test for both retrieval quality and prompt behavior. I used pprint to display the full structured output clearly, including the answer and source documents, so I can verify not just what the model said but also where that information came from.

```
In [123... pprint.pprint(qa_with_context("When did the IPL 2026 start?"))
```

```
{'query': 'When did the IPL 2026 start?',
 'result': 'IPL 2026 started on **28 March 2026**.',
 'source_documents': [Document(metadata={'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.', 'title': '2026 Indian Premier League', 'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League'}, page_content='The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.\n\n\n== Background ==\n\nThe Indian Premier League (IPL) is a professional Twenty20 (T20) cricket league, organized by the Board of Control for Cricket in India (BCCI). It has held annually since the first edition in 2008. Royal Challengers Bengaluru are the defending champions, having won their maiden title in the previous season after defeating Punjab Kings in the final.'),
 Document(metadata={'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League', 'title': '2026 Indian Premier League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.'}, page_content='The IPL Governing Council had initially announced that the IPL would expand to 84 matches for 2026 and 2027, with it expected to expand to 94 matches from 2028 onwards with the return of the complete double round-robin format that was used until 2021. However, the number of matches was kept as 74 for one more edition as it was in the previous four seasons.'),
 Document(metadata={'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.', 'title': '2026 Indian Premier League', 'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League'}, page_content='== External links ==\n\nOfficial website\n2026 Indian Premier League at ESPNcricinfo'),
 Document(metadata={'title': '2026 Indian Premier League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.', 'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League'}, page_content='==== Marketing ====\n\nThe Tata Group renewed their contract as the title sponsors of the IPL for a tenure of 5 years (2024–2028) for ₹2,500 crore (US$300 million).\n\n\n=== Broadcasting ===\n\nJioStar's Star Sports and JioHotstar, currently holds the satellite and digital broadcasting rights for the remainder of the 2023–2027 cycle respectively.\n\n\n\n== Teams ==\n\nThe same 10 teams from the previous season returned.\n\n\n\n=== Personnel changes ===")]]}
```

In this step, I run a second evaluation query to test whether the RAG system can retrieve and summarize a more nuanced topic, not just a simple date or match fact.

```
In [124... pprint.pprint(qa_with_context("What were the controversies regarding players in IPL 2026?"))
```



```
{'query': 'What were the controversies regarding players in IPL 2026?',
 'result': 'The player-related controversy mentioned for IPL 2026 was about '
           'Bangladeshi fast bowler Mustafizur Rahman.\n'
           '\n'
           '- He was bought by Kolkata Knight Riders for ₹9.2 crore in the '
           'auction.\n'
           '- After anti-Hindu violence in Bangladesh during 2025 and the '
           'lynching of Dipu Chandra Das in December 2025, there were calls to '
           'exclude Bangladeshi players from the IPL.\n'
           '- In January 2026, the BCCI asked the franchise to release '
           'Mustafizur Rahman from the squad and allowed them to choose a '
           'replacement.\n'
           '- This decision became controversial and drew mixed reactions:\n'
           '  - Criticized by Madan Lal, Shashi Tharoor, Khaled Mahmud, and '
           'Mohammad Ashraful.\n'
           '  - Defended by Aakash Chopra and BJP politician Sangeet Singh '
           'Som.\n'
           '\n'
           'So, the main players' controversy in IPL 2026 was the '
           'exclusion/release of Mustafizur Rahman, tied to political and '
           'communal tensions involving Bangladesh.',
 'source_documents': [Document(metadata={'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.'},
                                {'title': '2026 Indian Premier League'},
                                page_content='The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.\n\n\n== Background ==\n\nThe Indian Premier League (IPL) is a professional Twenty20 (T20) cricket league, organized by the Board of Control for Cricket in India (BCCI). It has held annually since the first edition in 2008. Royal Challengers Bengaluru are the defending champions, having won their maiden title in the previous season after defeating Punjab Kings in the final.'),
                                Document(metadata={'title': '2026 Indian Premier League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.'},
                                {'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League'},
                                page_content="This led the BCCI to put the Indian tour of Bangladesh on hold due to tensions. Furthermore, the Bangladesh Cricket Board requested the International Cricket Council (ICC) to shift their matches at the 2026 Men's T20 World Cup which was to be co-hosted by India and Sri Lanka, out of India, similar to that of Pakistan's. But the request was rejected by the ICC, and Bangladesh ultimately withdrew from the T20 World Cup. The Bangladeshi government also banned the broadcast of IPL in their country. Following the 2026 Bangladeshi general election, the newly elected government clarified that the broadcast of the IPL is not banned anymore in their country."),
                                Document(metadata={'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL 2026, is the 19th edition of the Indi
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an Premier League, a professional Twenty20 cricket league. The tournament features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Bengaluru are the defending champions.', 'title': '2026 Indian Premier League'}, page_content='The IPL Governing Council had initially announced that the IPL would expand to 84 matches for 2026 and 2027, with it expected to expand to 94 matches from 2028 onwards with the return of the complete double round-robin format that was used until 2021. However, the number of matches was kept as 74 for one more edition as it was in the previous four seasons.')

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In this step, I ask a targeted fact-check question to verify whether the RAG pipeline can retrieve a specific fixture-level detail from deep within the IPL 2026 page. This is a test to make sure RAG chain is working correctly and refusing to guess.

```
In [125... pprint.pprint(qa_with_context("Who won the 'Match 28' to be played on '19 April 2026' in the IPL 2026?"))
```

```
{'query': "Who won the 'Match 28' to be played on '19 April 2026' in the IPL "
'2026?',
'result': "I don't know.\n"
'\n'
'The context only provides general IPL 2026 tournament details, '
'scheduling background, and fixture announcement information. It '
'does not include the result of Match 28 on 19 April 2026.',
```

```
'source_documents': [Document(metadata={'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded a
s TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tourname
nt features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Chall
engers Bengaluru are the defending champions.', 'title': '2026 Indian Premier League', 'source': 'https://en.wikipedia.
org/wiki/2026_Indian_Premier_League'}, page_content='The 2026 Indian Premier League, also known as IPL 19 and branded a
s TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tourname
nt features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Chall
engers Bengaluru are the defending champions.\n\n\n== Background ==\n\nThe Indian Premier League (IPL) is a professional
Twenty20 (T20) cricket league, organized by the Board of Control for Cricket in India (BCCI). It has held annually sinc
e the first edition in 2008. Royal Challengers Bengaluru are the defending champions, having won their maiden title in
the previous season after defeating Punjab Kings in the final.'),
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Document(metadata={'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded a
s TATA IPL 2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tourname
nt features 10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Chall
engers Bengaluru are the defending champions.', 'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League',
'title': '2026 Indian Premier League'}, page_content='The IPL Governing Council had initially announced that the IPL wo
uld expand to 84 matches for 2026 and 2027, with it expected to expand to 94 matches from 2028 onwards with the return
of the complete double round-robin format that was used until 2021. However, the number of matches was kept as 74 for o
ne more edition as it was in the previous four seasons.'),
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Document(metadata={'title': '2026 Indian Premier League', 'source': 'https://en.wikipedia.org/wik
i/2026_Indian_Premier_League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL
2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features
10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Benga
luru are the defending champions.'}, page_content='In March 2026, it was announced that the fixtures for that season wo
uld be announced in two parts due to the State Assembly Elections in Assam, Tamil Nadu and West Bengal; for which the d
ates were not announced till 15 March. The first part of the schedule comprising the first twenty matches was announced
on 11 March, with the inaugural match being hosted at M. Chinnaswamy Stadium between the defending champions Royal Chal
lengers Bengaluru and Sunrisers Hyderabad on 28 March. The complete fixtures for the league stage were announced on 26
March. The playoffs schedule is yet to be announced.'),
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Document(metadata={'source': 'https://en.wikipedia.org/wiki/2026_Indian_Premier_League', 'title':
'2026 Indian Premier League', 'summary': 'The 2026 Indian Premier League, also known as IPL 19 and branded as TATA IPL
2026, is the 19th edition of the Indian Premier League, a professional Twenty20 cricket league. The tournament features
10 teams competing in 74 matches. It is being played from 28 March to 31 May across 13 venues.\nRoyal Challengers Benga
luru are the defending champions.'}, page_content="=== Schedule ===\n\nAccording to ESPNcricinfo, the dates of the tourna
ment window for the three seasons between 2025 and 2027 were sent to franchises ahead of the 2025 auction. In November
2024, the tentative dates for the three seasons were announced, with the 2026 edition set to take place from 15 March t
o 31 May. In December 2025, it was announced that the 2025 season would commence on 26 March to allow for a three-week
```

gap after the 2026 Men's T20 World Cup final scheduled to take place on 8 March. In February 2026, it was announced that the 2026 season would commence 2 days later on 28 March. This led to a clash with the 2026 Pakistan Super League for the second consecutive year taking place from 26 March to 3 May 2026, which was again moved from its typical February–March window for the same reason."]]}

In this diagnostic step, I directly check whether the phrase Match 28 exists in the chunked documents before retrieval. The first line returns a boolean to confirm presence at least once, and the second line returns the exact number of chunks containing that phrase. This helps isolate whether the issue is in ingestion and chunking or in retrieval ranking: as the result is False and 0, the relevant section was never indexed.

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In [126... print(any("Match 28" in d.page_content for d in data))  
print(sum("Match 28" in d.page_content for d in data))
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False

0